

Kaipeng Li

Master's Student and Researcher

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Summary

Master's student and independent researcher with expertise in recommender systems, information retrieval, and machine learning. Passionate about advancing the frontiers of cross-modal retrieval and LLM-based applications.

Education

Mas- ter of	University of Tsukuba , Computer Science • GPA: 3.6/4.3 • Focus on multimodal retrieval and generative models	Tsukuba, Japan Oct 2023 – present
Sci- ence		
Research Stu- dent Pro- gram	University of Tsukuba , Computer Science • Participated in advanced research projects	Tsukuba, Japan Apr 2022 – Sept 2023

Experience

Tsukuba Technology Co. Ltd , Software Development Engineer Intern Developing image processing systems for security inspection machines and customs applications. • Participated in image processing systems development for security inspection machines • Optimized software user interface and integrated logging/data tracking features • Developed Modbus-compatible software systems for real-time hardware monitoring • Languages: C, C++, Python, OpenCV	Tsukuba, Japan Feb 2024 – present 2 years 5 months
Computational Intelligence Laboratory, UESTC , Research Assistant Assisted in research on latest technology trends and international conferences. • Investigated international top conferences and technical references • Prepared research reports and project proposals • Provided technical solutions and promoted project implementation	Chengdu, China July 2021 – Mar 2022 10 months
Sichuan Huadi Information Technology Co., Ltd. , Embedded Software Development Intern Developed Qt-based video surveillance software under Linux. • Implemented multi-speed playback function for video search optimization • Developed Qt-based video surveillance software for real-time area monitoring • Created Qt program using V4L2 driver for camera real-time monitoring • Languages: C, C++, Python, Qt	Chengdu, China June 2020 – July 2020 3 months

Publications

A Combination-based Framework for Generative Text-image Retrieval: Dual Identifiers and Hybrid Retrieval Strategies A framework for generative text-image retrieval using dual identifiers and hybrid retrieval strategies. Kaipeng Li, Haitao Yu, Yubo Fang, Chao Lei 10.1145/3767695.3769474 (Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval)	Dec 2025
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FlowCoPCL: Enhanced Flow Correlation Attacks on Tor Using Patching and Contrastive Learning

Apr 2026

A study on flow correlation attacks on Tor using patching and contrastive learning.

Shiping Huang, Weina Niu, Ruoyu Wang, Tianlang Zhang, *Kaipeng Li*, Yanfeng Zhang, Xiaosong Zhang

[10.26599/TST.2024.9010165](https://arxiv.org/abs/10.26599/TST.2024.9010165) (Tsinghua Science and Technology)

Skills

Programming Languages: C, C++, Python, Java, JavaScript, SQL

Machine Learning: TensorFlow, PyTorch, Deep Learning, Information Retrieval, Recommender Systems

Full Stack Development: MERN Stack, MongoDB, Express.js, React, Node.js, Web Development

Software Development: Image Processing, OpenCV, Qt, Embedded Systems, Modbus

Data Analysis: XGBoost, Transformer Models, Statistical Analysis, Data Visualization

Languages

Chinese (Mandarin): Native speaker

English: Fluent

Japanese: Intermediate

Interests

Research: Text-image retrieval, multimodal learning, recommender systems, generative AI

Projects

Multimodal Retrieval Research

Research on cross-modal retrieval, generative retrieval models, and retrieval-augmented generation.

- Cross-modal text-image retrieval
- Biomedical generative retrieval
- Learning-to-rank loss generation